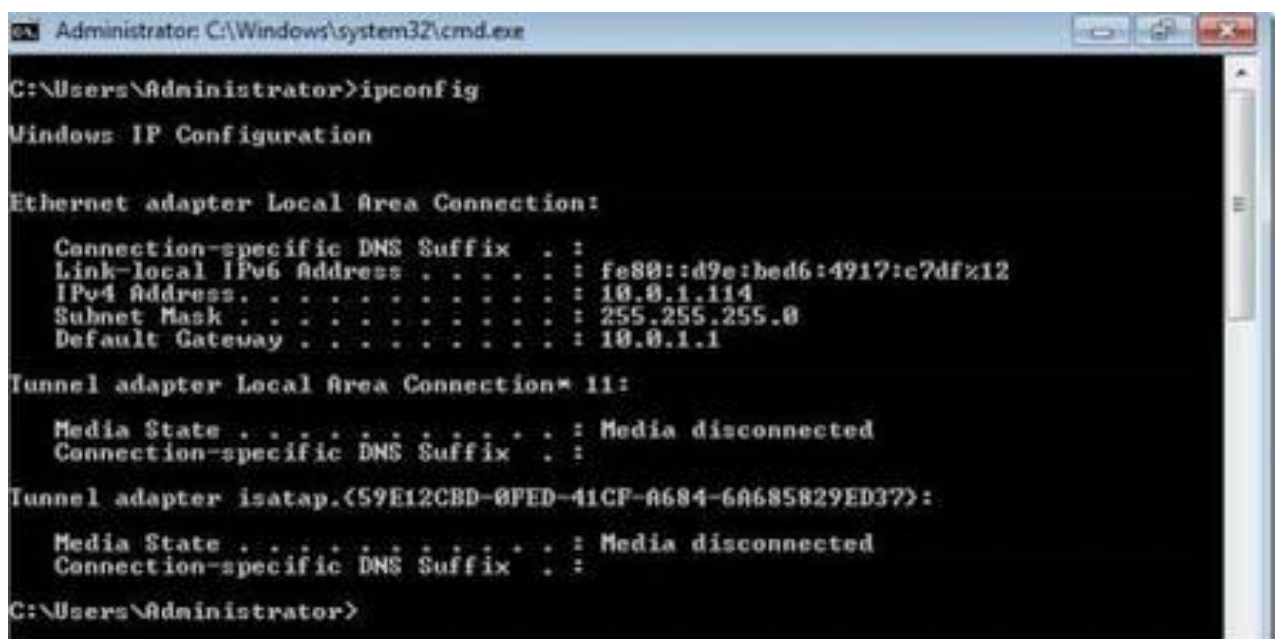


Configure IPv6 Addressing

Task 1 :- Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

ANS :-



```
Administrator: C:\Windows\system32\cmd.exe

C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::d9e:bed6:4917:c7df%12
    IPv4 Address. . . . . : 10.0.1.114
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.1.1

Tunnel adapter Local Area Connection* 11:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter isatap.{59E12CBD-0FED-41CF-A684-6A685829ED37}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\Administrator>
```

Task 2 :- Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

ANS :-

```
C:\Users\>ping fs6712x

Pinging FS6712X.local [2600:8800:6209:f400: ] with 32 bytes of data:
Reply from 2600:8800:6209:f400:7a72: : time=1ms
Reply from 2600:8800:6209:f400:7a72: : time=3ms
Reply from 2600:8800:6209:f400:7a72: : time=1ms
Reply from 2600:8800:6209:f400:7a72: : time=3ms

Ping statistics for 2600:8800:6209:f400:7a72: :
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

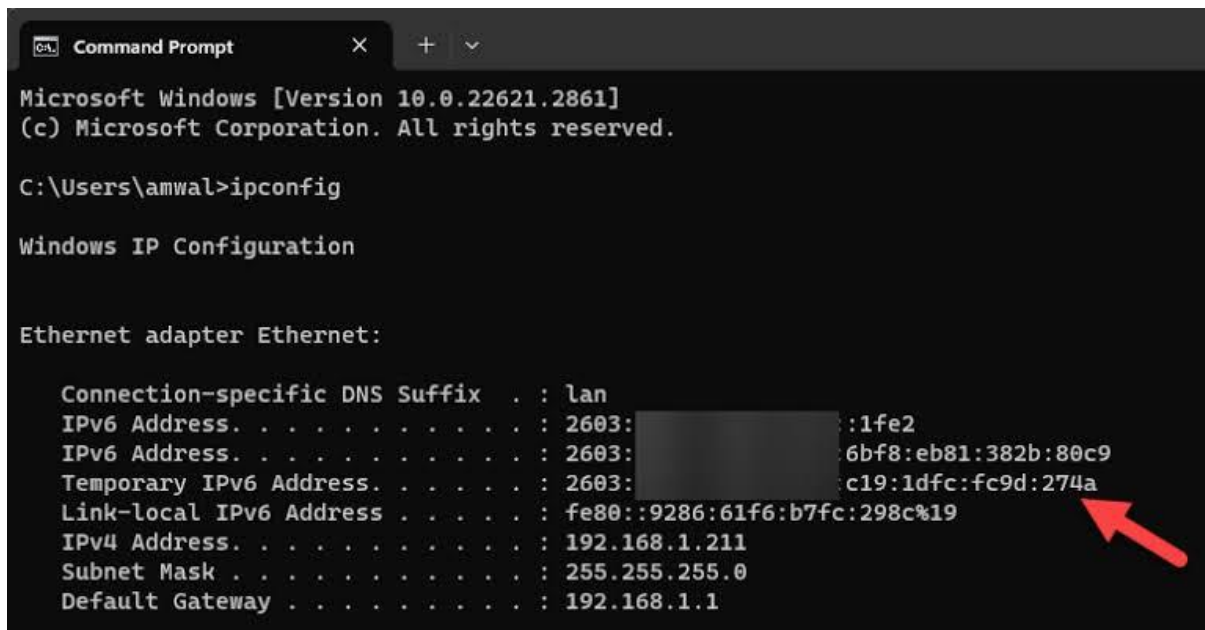
C:\Users\>ping -4 fs6712x

Pinging FS6712X.local [192.168.0.105] with 32 bytes of data:
Reply from 192.168.0.105: bytes=32 time=1ms TTL=64
Reply from 192.168.0.105: bytes=32 time=1ms TTL=64
Reply from 192.168.0.105: bytes=32 time=1ms TTL=64
Reply from 192.168.0.105: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.0.105:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms
```

Task 3 :- Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

ANS :-

A screenshot of a Windows Command Prompt window. The title bar says 'Command Prompt'. The text inside shows the command 'ipconfig' being executed. The output displays network configuration for an Ethernet adapter. It lists several IPv6 addresses: a standard IPv6 address, a link-local IPv6 address, and a 'Temporary IPv6 Address'. A red arrow points to the 'Temporary IPv6 Address' line, which is '2603:c19:1dfc:fc9d:274a'. The IPv4 address is 192.168.1.211 and the default gateway is 192.168.1.1.

```
Microsoft Windows [Version 10.0.22621.2861]
(c) Microsoft Corporation. All rights reserved.

C:\Users\amwal>ipconfig

Windows IP Configuration

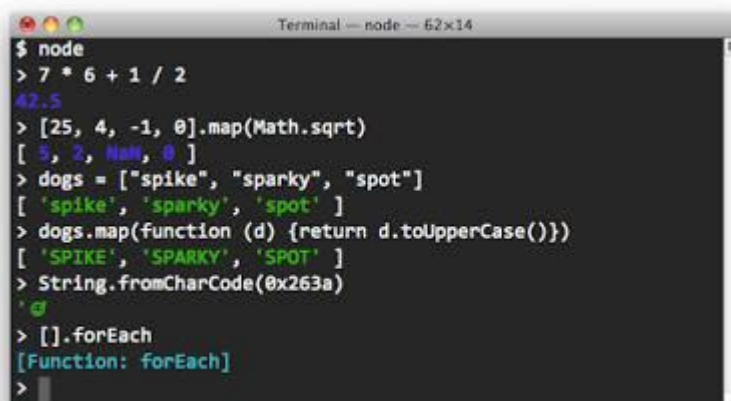
Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : lan
    IPv6 Address. . . . . : 2603: [redacted] :1fe2
    IPv6 Address. . . . . : 2603: [redacted] :6bf8:eb81:382b:80c9
    Temporary IPv6 Address. . . . . : 2603: [redacted] :c19:1dfc:fc9d:274a
    Link-local IPv6 Address . . . . . : fe80::9286:61f6:b7fc:298c%19
    IPv4 Address. . . . . : 192.168.1.211
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
```

Notice the Temporary IPv6 Address entry. When privacy extensions are active, outgoing internet connections automatically prefer this temporary address over your main static or dynamic SLAAC address. Upon a system reboot or reconnecting to the network, this randomized interface identifier updates automatically.

Task 4 :- Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

ANS :-

A terminal window titled "Terminal — node — 62x14" showing the execution of a Node.js script. The prompt is "\$ node". The user enters "> 7 * 6 + 1 / 2", and the output is "42.5". The user enters "> [25, 4, -1, 0].map(Math.sqrt)", and the output is "[5, 2, NaN, 0]". The user enters "> dogs = [\"spike\", \"sparky\", \"spot\"]", and the output is "['spike', 'sparky', 'spot']". The user enters "> dogs.map(function (d) {return d.toUpperCase()})", and the output is "['SPIKE', 'SPARKY', 'SPOT']". The user enters "> String.fromCharCode(0x263a)", and the output is "'🐶'". The user enters "> [].forEach", and the output is "[Function: forEach]". The prompt is ">".

```
$ node
> 7 * 6 + 1 / 2
42.5
> [25, 4, -1, 0].map(Math.sqrt)
[ 5, 2, NaN, 0 ]
> dogs = ["spike", "sparky", "spot"]
[ 'spike', 'sparky', 'spot' ]
> dogs.map(function (d) {return d.toUpperCase()})
[ 'SPIKE', 'SPARKY', 'SPOT' ]
> String.fromCharCode(0x263a)
'🐶'
> [].forEach
[Function: forEach]
>
```

How to run it:

1. Copy the code into a file named search.js.
2. Run it in your terminal: node search.js
3. Type a partial name (e.g., bir or king) and hit Enter.

Task 5 :- Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

ANS :-

Summary of Document Content :-

1. AI-Generated Subnetting Scenario :-

- Allocated Parent Block: 2001:db8:abcd::/48
- Subnet 1 (Corporate HQ & Management): 2001:db8:abcd:1::/64 (Gateway: 2001:db8:abcd:1::1)
- Subnet 2 (Research & Engineering): 2001:db8:abcd:2::/64 (Gateway: 2001:db8:abcd:2::1)
- Subnet 3 (Guest Wi-Fi & DMZ): 2001:db8:abcd:3::/64 (Gateway: 2001:db8:abcd:3::1)

2. Selected Assignment & Configuration Steps :-

- Selected Subnet: Subnet 2 (2001:db8:abcd:2::/64)
- Assigned Host Address: 2001:db8:abcd:2::100/64

IPv6 SUBNETTING SCENARIO & CONFIGURATION REPORT

=====

PART 1: AI-GENERATED IPv6 SUBNETTING SCENARIO

[Prompt provided to AI]:

"Generate a sample IPv6 subnetting scenario for a medium-sized enterprise network.

Allocate a /48 prefix and create at least 3 distinct subnets (/64 each) with clear

departmental assignments and usable IP ranges."

[AI Response / Scenario Breakdown]:

Parent Allocated Block: 2001:db8:abcd::/48

Subnet Allocations (/64 subnets):

1. Subnet 1: Corporate HQ & Management

- Subnet ID: 2001:db8:abcd:0001::/64
- Subnet Range: 2001:db8:abcd:0001:0000:0000:0000:0000
to 2001:db8:abcd:0001:ffff:ffff:ffff:ffff
- Gateway: 2001:db8:abcd:1::1

- Purpose: Internal server infrastructure & core management devices

2. Subnet 2: Research & Engineering

- Subnet ID: 2001:db8:abcd:0002::/64

- Subnet Range: 2001:db8:abcd:0002:0000:0000:0000:0000
to 2001:db8:abcd:0002:ffff:ffff:ffff:ffff

- Gateway: 2001:db8:abcd:2::1

- Purpose: Development workstations, build servers, lab environment

3. Subnet 3: Guest Wi-Fi & DMZ

- Subnet ID: 2001:db8:abcd:0003::/64

- Subnet Range: 2001:db8:abcd:0003:0000:0000:0000:0000
to 2001:db8:abcd:0003:ffff:ffff:ffff:ffff

- Gateway: 2001:db8:abcd:3::1

- Purpose: Visitor Wi-Fi access and public-facing isolated devices

PART 2: MANUAL CONFIGURATION STEPS

Selected Subnet for Configuration: Subnet 2 (Research & Engineering)

Assigned System IPv6 Address: 2001:db8:abcd:2::100/64

Default Gateway Address: 2001:db8:abcd:2::1

--- OPTION A: WINDOWS CONFIGURATION ---

1. Open Command Prompt or PowerShell as Administrator.

2. Verify network interface name:

> netsh interface ipv6 show interfaces

3. Assign the valid IPv6 address from Subnet 2 to primary adapter (e.g., "Ethernet"):

> netsh interface ipv6 add address interface="Ethernet" address="2001:db8:abcd:2::100/64"

4. Configure the default gateway:

> netsh interface ipv6 add route prefix="::/0" interface="Ethernet" nexthop="2001:db8:abcd:2::1"

5. Verify assignment and connectivity:

> ipconfig /all

> ping -6 2001:db8:abcd:2::1

--- OPTION B: LINUX CONFIGURATION ---

1. Open Terminal with root or sudo privileges.

2. Identify network interface name:

> ip link show

3. Assign the valid IPv6 address from Subnet 2 to interface (e.g., "eth0"):

\$ sudo ip -6 addr add 2001:db8:abcd:2::100/64 dev eth0

4. Configure default IPv6 gateway route:

\$ sudo ip -6 route add default via 2001:db8:abcd:2::1 dev eth0

5. Verify assigned address and network routing:

\$ ip -6 addr show dev eth0

\$ ping -6 2001:db8:abcd:2::1

=====
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END OF REPORT & CONFIGURATION

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